Migration high level outline
How will you initiate a migration process?

- there are 4 key applications
- two are P1
- one is P2
- another one is P3
- You have one week to complete migration planning
- You can ask any question until tomorrow evening

1. What is the current setup (Assessment)
    - Technical assessment (involve Sr. Technical)
        - on-premise
        - on-cloud (other than AWS)
        - how many applications are there with you (4/10)
    - Financial Assessment (current OpEx)
    - Security Assessment (3rd party tools used, license, cloud compatibility)
    - Compliance & Regulatory Assessment (involve legal and IT audit teams)
    - Technical and Functional assessment (involve business representative)
2. Understand Business and Technology Objectives of migrating Applications.
    - who are the stakeholders
        - internal and external (both)
    - how does this impact revenue generation
        - (for example, 40% of the revenue related to migrating applications)
    - why is an application using this technology (on-premises)
    - what are budget constraints
    - Capture current OpEx for each application
3. End to End Application workflow
    - other connected or participating applications will be revealed in this process
        - upstream and downstream both
        - find and document the mode of interface with each of these applications
    - User Roles currently in practice
    - how do they use these apps (mobile, desktop thick client...)

4. Application-wise Regulatory requirements
    - consider all applicable guidelines that have influence on the migrating applications
    - Regional restrictions for data movement (specific type of data or all)

(You may consider building a Relational Database for following information, or
connected excel sheets)

5. For each Applications find following
    - 5.0 Application name & description
    - 5.1 Development technology (.Net, Java, etc.)
    - 5.2 Third party tools used
        - 5.2.1 Third party license agreements
        - 5.2.2 Support for these tools in cloud environment
        - 5.2.3 Get a head start by sending an email to third party about your plan to migrate and request for temp license so that you can enable cloud testing initially
        - 5.2.4 If there is a different license required for cloud then start procurement process
    - 5.3 How many servers for these applications are in use currently (See Pt 7.)
    - 5.4 Upstream systems this application depends on (list all those applications)
    - 5.5 How many distinct components make up this application (for list of those components see Pt 6.)
    - 5.6 How many database (different schemas) used by this application
    - 5.7 Number of Batch Job(s) executed against this application and their significance (hourly, daily, monthly, etc.)
    - 5.8 What reports are generated and who are the recipients
        - 5.8.1 How are these reports circulated. Email, intranet, Sharepoint, etc.
        - 5.8.2 Report format used. List all formats used.

5.8.3 Are historical reports retained somewhere, if yes, how and where.

5.9 Business owner name, primary and secondary

5.10 Technical owner name, primary and secondary

5.11 Escalation matrix for both business side and technical side

5.12 Apply 7R + mix

(mix is used when more than one component will undergo change for Re-Platforming or Re-Architect)

5.13 Issues encountered and lessons learned (consider past 6 months to 1 year)

5.14 User authentication and authorization mechanism

- SSO enabled (yes/no)

(for SSO, consider interface with Identity Provider (LDAP/AD) as top priority)

6. For each application components find out
(File Server, Web Server, Application Server, API module, UI modules, Cron Job, Report, etc.)

6.0 Component name and description

6.1 Inputs received from which component(s) (one or more)

6.2 Interface formats (API, batch, etc.)

6.3 Output produced by this component with formats

(there can be more than one mode and format)

6.4 Hosted on - Server ID

6.5 Apply 7R

6.6 Issues encountered and lessons learned (past 6 months minimum)

6.7 Special Libraries and configuration

7. For each Server used by the applications being migrated

7.0 Name of the server and configuration details

(CPU cores, Hz, memory, boot drive size, OS & version,
attached storage, network interface card(s), any special hardware* used, etc.)

7.1 Operating System version

- If cloud has latest OS version, first test compatibility,
do a POC in cloud before planning or find out details from respective vendor

7.2 Physical machine, VMWare or Hyper-V?

7.3 List of scheduled tasks with frequency and relationship with applications being migrated

7.4 List all licensed software used (including OS and third-party s/w)

7.5 Directly related to migrating application(s) or indirectly related

7.6 Take backed-up up version of the server and restore on some other machine to check backup quality. (this is important & critical too)

7.7 Apply 7R + mix

7.8 Issues encountered and lessons learned

8. Database(s) used by the migrating Applications

8.1 Database vendor and version (Oracle 11g, MS SQL 2016, MySQL 8, etc.)

8.2 JDBC or ODBC driver version

8.3 For each database

8.3.1 Number of tables

8.3.2 Current row count (as of date)

8.3.3 Monthly growth %

8.3.4 Primary key, Foreign key and other constraints if any. ERD would be better.

8.3.5 List of Views, Functions, Stored Procedures

8.3.6 Database size (as of date)

8.3.7 Current Backup and storage process/procedure

8.3.8 Current Restore process/procedure

8.3.8.1 Check backup quality and reliability

(conduct this test before starting the migration process)

8.4 Apply 7R + mix

(mix here would mean, along with target DB, you would also use S3, RedShift, Storage Gateway, etc.)

8.5 Issues encountered and lessons learned

8.6 Endianness and date/time format conflict or time zone issues for migration

(critical for migration changing platforms, from Solaris, AIX, etc) https://geraldonit.com/2017/09/04/big-and-little-endian-operating-systems/

9. For each point above starting from see pt. 5 (including subpoints)
   consider 7R and update that column as appropriate.
   9.1 AWS architect or consultant (it can be you too) should help you define these.
   9.2 To-Be Architecture must be ready and approved by stakeholders before you
   finalize 7R assessment to each line item above.
10. When you are ready for migration planning phase
   10.1 Consider the logistics involved in migrating Server,
   Application components and test data, test plans.
   10.2 Conduct an end to end test to ensure you have migrated all application components
   correctly as the plan and they are functioning as intended.
   10.2.1 Do not focus on performance in this phase.
   Just focus on applications end to end functionality
   10.3 Fix issues that you may encounter and document these in appropriate line item
   10.4 Redo end to end deployment.
   (if you can use CloudFormation at this point in time then do use it.
   Use Cloud Former feature to generate CloudFormation code for you and
   tweak as appropriate)
   10.5 For database part, migrate only 10% to 20% data for conducting test.
   Whilst application is being tested for functionality, database records can be
   tested for completeness and correctness.
   10.6 Largely DMS would take care of data migration either directly via console or
   indirectly via SCT for schema conversion.
   Strongly recommend Direct-Connect or VPN that will help you do one-time data migration.
   Post one-time data migration, CDC (Change Data Capture) can be employed
   via DMS or other tools.
   10.7 Use Well-Architecture Tool and answer all the questions.
   This will help you plan for Security, IAM policies, employing AWS Config,
   CloudTrail, VPC Flow logs, AWS WAF, etc. as appropriate.

It is strongly recommended that you adopt PESSIMISTIC approach at every stage and plan for remediation.
have plan B and C ready and define when to activate which plan.
This will help you address any eventual challenges
Ensure things are under control.

With reference to our use case...
Do not be surprised that you were to migrate only two P1 applications, basis its dependencies you
may end up migrating all 10 applications.
Business folks do not necessarily understand all the technical
part from application standpoint.

Plan a POC with representative data for all Applications and related components.
- Identify all related components correctly
- helps with migration time estimates (guestimation)
- performance requirements
- tools identification
- ETL process if applicable

After migration is complete, think about
Audit requirements,
Monitoring requirements,
Alerts and Event notifications
Backup and Restore - conduct restore drills to ensure they work and can be relied upon.
Cloud environments do not require DR plans due to its inherent nature of HA, resiliency, etc.
Cross region DR setup is still prevailing though, mostly for data backups.
Cloud Opex, performance and security related
1. Run your system for at least three months to understand the usage and spends
2. First evaluate the spending per component and aggregate it to application level.
3. Find ways to optimize the cost, mostly EBS, EFS and data flowing out of cloud. Be frugal.
   enable compression where possible. (S3, Web Data, etc.)

4. When optimizing for cost change only one thing at a time for a given month to ensure you
have better control and visibility into what is causing excess spending.
Have a clear end date defined for Direct-Connect, VPN, Storage Gateway, etc.
unless they are employed as an ongoing strategic requirement.
5. Ensure you are not compromising performance for cost, or HA for cost, etc.
6. Periodically review security posture, grant least privileges all the time,
consider using roles first, employ cloud enabled services to protect
applications and resources.

Few useful references for migration activity see cloud KPI checklist here

https://blog.newrelic.com/engineering/cloud-migration-checklist/
https://d1.awsstatic.com/whitepapers/migrating-oracle-database-workloads-to-oracle-linux-on-aws.pdf
https://docs.aws.amazon.com/fsx/latest/WindowsGuide/data-protection.html

Physical Server Migration Option via third-party

https://www.rivermeadow.com/blog/migrate-physical-servers-to-aws

Following link is a good reference to understand various ways to deploy MS-SQL with High Availability (HA) can be used on AWS

https://docs.cloud.oracle.com/en-us/iaas/Content/Resources/Assets/whitepapers/deploy-sql-server-availability-groups.pdf
https://d0.awsstatic.com/whitepapers/Storage/AWS%20Storage%20Services%20Whitepaper-v9.pdf

Shrikant G Jannu (join me at LinkedIn, feel free to ask any questions you may have)
www.linkedin.com/in/shrikantgjannu